

Speak to a Protein: An Interactive Multimodal Co-Scientist for Protein Analysis

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Abstract

Building a working mental model of a protein typically requires weeks of reading, cross-referencing crystal and predicted structures, and inspecting ligand complexes, an effort that is slow, unevenly accessible, and often requires specialized computational skills. We introduce *Speak to a Protein*, a new capability that turns protein analysis into an interactive, multimodal dialogue with an expert co-scientist. The AI system retrieves and synthesizes relevant literature, structures, and ligand data; grounds answers in a live 3D scene; and can highlight, annotate, manipulate and see the visualization. It also generates and runs code when needed, explaining results in both text and graphics. We demonstrate these capabilities on relevant proteins, posing questions about binding pockets, conformational changes, or structure-activity relationships to test ideas in real-time. *Speak to a Protein* reduces the time from question to evidence,

lowers the barrier to advanced structural analysis, and enables hypothesis generation by tightly coupling language, code, and 3D structures. *Speak to a Protein* is freely accessible at ([URL to be disclosed after publication](#)).

1 Introduction

Proteins are the molecular machinery of life, and understanding their structure and function is fundamental to modern biology and medicine. For a researcher in drug discovery or molecular biology, developing an intuitive, "working mental model" of a target protein, its active sites, its conformational dynamics, and its network of interactions is a critical first step. However, this process is slow and arduous, often requiring a combination of deep domain knowledge and specialized computational skills.

A researcher investigating a protein kinase to understand how a new series of inhibitors might bind must embark on a fragmented and technically demanding workflow. This involves sifting through the PubMed literature¹, fetching and comparing multiple structures from the Protein Data Bank (PDB)², querying UniProt³ for functional annotations and disease-associated variants, and extracting structure-activity relationship (SAR) data from databases like ChEMBL⁴. Each step requires navigating different interfaces and data formats. Furthermore, deeper analysis, such as superimposing structures, identifying key interactions, or plotting bioactivity data, requires proficiency with specialized software or scripting languages, creating a significant barrier to entry for many bench scientists. This high friction for asking and answering questions restrains curiosity and slows the pace of discovery.

To address these challenges, we introduce *Speak to a Protein*, a new capability designed to transform protein analysis into an interactive dialogue with an AI co-scientist that collaborates with the user in real-time. Using recent advances in large language models (LLMs) and their evolution into multimodal scientific agents⁵⁻⁷, our system can comprehend complex, natural language queries about a protein of interest. It autonomously retrieves, integrates,

and synthesizes information from a comprehensive suite of biological data sources, including literature, structural repositories, and biochemical databases.

The core innovation of *Speak to a Protein* is its ability to ground its responses across multiple, synchronized modalities. When a user asks a question, the AI does not simply return text. It interacts with a live 3D structural viewer to highlight residues, measure distances, or annotate binding pockets. It can generate and execute code in a sandboxed environment to perform calculations, filter tabular data, or generate plots on the fly. Furthermore, the AI co-scientist sees, understands and controls the visualization, offering a natural interaction with the user. This tight coupling of natural language, 3D visualization, and code execution creates a seamless and intuitive environment for scientific exploration.

This paper makes the following contributions: We present the design and architecture of an AI system that integrates language, code execution, and 3D visualization for interactive protein analysis. We show how this multimodal approach drastically lowers the barrier to complex structural and biochemical data analysis. Through case studies on relevant proteins and benchmarks, we show that this system enables a more fluid and powerful form of hypothesis generation, accelerating the cycle from question to evidence.

2 Related Work

The ambition to create an AI capable of scientific discovery is a long-standing goal, articulated in visions such as Hiroaki Kitano’s proposal for an “AI Scientist”: a system that could autonomously formulate hypotheses, design experiments, and achieve Nobel-class discoveries⁸. While such a fully autonomous system^{9,10} remains a grand challenge, recent progress in large language models (LLMs) has enabled the development of a more immediate and collaborative paradigm: the “AI co-scientist” or “advanced intelligence” in Kitano’s words.

Early systems explored conversational interfaces for structural inspection and Q&A over proteins. Guo et al.¹¹ demonstrated *ProteinChat*, which couples LLM prompting with pro-

tein 3D structures to answer user questions about residues and pockets. Contemporary efforts such as Wang et al.¹² and Xiao et al.¹³ investigate protein-aware prompting and multimodal conditioning for function/property reasoning. Most recently, Wang et al.¹⁴ proposes *Prot2Chat*, an LLM that fuses protein sequence, structure, and text via an early-fusion adapter, directly targeting protein Q&A.

Beyond text-only chat, domain copilots increasingly drive molecular viewers and modeling tools. Sun et al.¹⁵ introduce *ChatMol Copilot*, an LLM agent that coordinates cheminformatics and modeling tools (e.g., docking, conformer generation) in response to natural-language requests. In parallel, Ille et al.¹⁶ systematically evaluates GPT-4’s ability to perform rudimentary structural modeling and protein–ligand interaction analysis, highlighting both promise and limitations. Our work is aligned with this line but centers on tightly coupling language, code execution, and a live 3D scene for grounded, interactive answers.

Agent frameworks augment LLMs with tool use, retrieval, and planning. *ChemCrow*¹⁷ shows that equipping GPT-4 with chemistry tools enables multi-step synthesis planning and materials tasks. More recently, CLADD¹⁸ proposes a retrieval-augmented multi-agent system specialized for drug discovery tasks. *Speak to a Protein* adopts the agentic paradigm for structural biology: it retrieves literature/structures, executes analyses (e.g., pocket mapping, SAR tables), and grounds responses in synchronized 3D visualizations.

Compared to prior work, our contribution is an end-to-end, *interactive* co-scientist for proteins that (i) unifies literature/structure/ligand retrieval, (ii) reasons with tabular and 3D modalities, (iii) executes code for analyses, and (iv) directly annotates/manipulates the 3D scene in response to dialogue (Table 1). This tightly coupled language–code–3D loop reduces the time from question to evidence relative to agent-only or text-only systems.

Table 1: Comparison of Speak to a Protein with representative LLM-based systems developed for drug discovery, as well as with general-purpose AI models. Speak to a Protein uniquely combines access to multiple biological databases, scientific literature extraction, on-the-fly code execution, real-time 3D molecular reasoning, and persistent knowledge management. These capabilities are not collectively available in other existing platforms

System	API	Lit. Extr.	SandBox	3D Reasoning	Memory
SpeakToAProtein	Yes	Yes	Yes	Yes	Yes
General AI	No	Yes	Yes	No	No
ProteinChat	No	No	No	No	No
Prot2Chat	No	No	No	Yes	No
ChatMolCopilot	No	No	No	Yes	No
ChemCrow	Yes	Yes	Yes	No	Yes
CLADD	Yes	Yes	No	No	No

3 System Overview: *Speak to a Protein*

3.1 Architecture

The system architecture of *Speak to a Protein* (Figure 1) is organized around four components: a frontend for user interaction and visualization, an agent harness that runs the agent loop and exposes tools to the language model, a search service that maintains semantic indices over codebases, documentation, and literature, and a sandboxed filesystem where code executes against a scientific Python runtime.

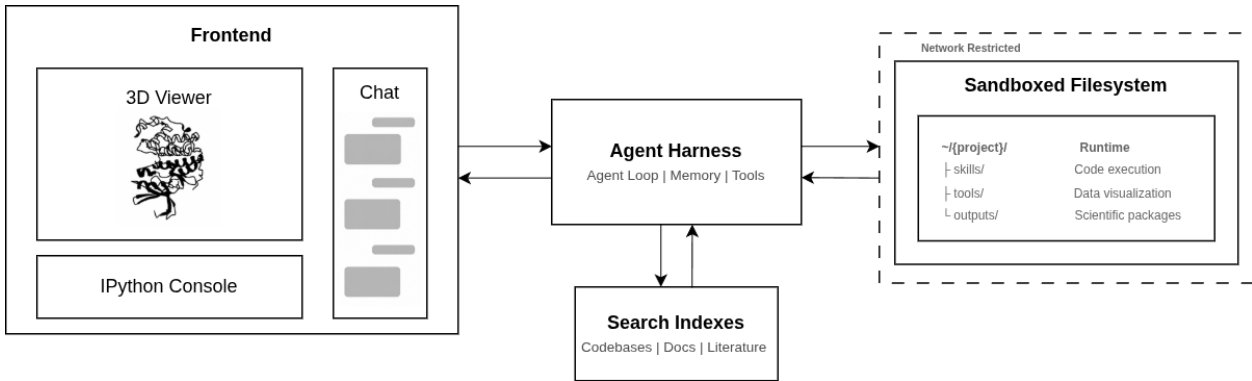


Figure 1: **Overview of the system architecture.** The Frontend hosts a chat, a 3D molecular viewer, and an in-browser IPython console. The Agent Harness runs the agent loop, holds conversation memory, and calls tools. A search service indexes codebases, documentation, and literature. A sandboxed filesystem provides a per-project workspace and a scientific Python runtime, with restricted network egress.

The frontend incorporates a conversational chat panel and a 3D molecular viewer (Figure 2 and Section 3.3). Alongside these, an in-browser IPython console powered by Pyodide allows users to inspect or extend any analysis directly in the browser, with no installation. The frontend renders the agent’s responses, displaying text in chat and applying viewer commands as the agent issues them.

The agent harness runs an agent loop that interleaves model reasoning with tool calls and maintains a per-session conversational memory, allowing the assistant to plan multi-step workflows and revisit earlier results. The underlying language model can run via an external API or as a self-hosted local model, with the choice made at configuration time. The harness exposes a small, general-purpose tool set:

- **Viewer Tools:** Issue commands to the 3D viewer to load structures, define selections, and produce visual manipulations.
- **Code Execution:** Run bash and Python commands inside the sandboxed filesystem.
- **File Operations:** Read, write, and edit files within the sandboxed filesystem.
- **Semantic Search:** Query the search service for passages from indexed library codebases (MoleculeKit¹⁹ and related packages), curated database documentation, and protein-conditioned literature corpora built on demand from PubMed Central (Section 3.2).
- **Sub-agents:** Delegate self-contained subtasks to spawned agents to keep the main context window concise.

Rather than wrapping each external scientific database in a dedicated tool, the system relies on *skills*: lightweight, on-demand instructions that teach the agent how to interact with public scientific resources. Each skill is a Markdown document, optionally accompanied by example scripts, describing access patterns, identifier conventions, rate limits, and idiomatic queries for a given resource. Skills are surfaced through progressive disclosure: at session start the agent receives a one-line description of every available skill, and when one

becomes relevant the agent reads the corresponding `SKILL.md` and uses it to write Python or bash inside the sandbox. We currently provide skills for protein and structure databases (UniProt, PDB, AlphaFoldDB), chemical and bioactivity databases (ChEMBL, PubChem, DrugBank, ZINC), gene and genomic databases (NCBI Gene, Ensembl), and biomedical literature retrieval (PubMed). New databases or workflows can be added by authoring a new skill, without modifying the core system.

The sandboxed filesystem is a Docker container with restricted egress, where outbound traffic is filtered by a forward proxy that allowlists scientific domains. The agent operates over a per-session project workspace alongside a pre-installed scientific Python runtime including MoleculeKit, RDKit, BioPython, and pandas. This colocation makes data retrieval and downstream analysis a single continuous activity: a skill-driven query against, for example, ChEMBL writes a CSV into the workspace, which the agent then loads with pandas in the same sandbox to filter compounds, compute statistics, or feed values into the viewer. Files generated by the agent persist across turns, supporting iterative refinement and full reproducibility of the underlying analyses.

3.2 Literature Search and Comprehension

The literature pipeline assembles a protein-conditioned corpus on demand and serves it as one of the indices behind the search service. While discovery relies on *curated references from UniProt*, the pipeline accepts a UniProt accession, a PDB identifier, or a FASTA sequence as input. For a UniProt accession, the references linked to that entry are augmented by expanding the selection to all PDB structures cross-referenced from it. For a PDB code, the system first queries UniProt for all entries cross-referenced to that structure, and then collects their reference lists. For a raw FASTA sequence, the system searches the Protein Data Bank to identify matching structures, resolves them to UniProt entries, and again retrieves their references. This pathway ensures that the system always incorporates the expert-curated literature that UniProt associates with a protein, regardless of the initial

identifier type.

Discovery yields a diverse set of article identifiers, including PubMed IDs, DOIs, and PubMed Central IDs. To unify them, we normalize all entries to *PubMed Central IDs* (*PMCID*s) using a public conversion service. Only the subset of publications openly available on PubMed Central can be downloaded and processed further; articles behind paywalls remain indexed only by their identifiers. For each accessible PMCID, we retrieve the article in XML format, which is efficient to fetch and preserves section and paragraph boundaries. The text is cleaned and segmented into coherent passages that are embedded into a vector space for retrieval-augmented generation (RAG) using LlamaIndex²⁰.

All passages for a given protein are combined into a *protein-conditioned retrieval index*, together with metadata such as PMCID, DOI, and the set of matched PDB and UniProt identifiers (Table S1). The index is cached on disk along with the associated protein identifiers, so future queries on the same target reuse the prebuilt corpus without repeated downloads. At query time, the system formulates a descriptive retrieval prompt, retrieves the top- k relevant passages, and returns them along with their citations. The retrieved text is provided to the language model as grounded context, either to directly answer a user’s question (e.g., ”Which mutations in CDK2 affect inhibitor binding?”) or to supply background for further analysis and additional tool calls (e.g., filtering ChEMBL assays for compounds tested in the cited studies, or highlighting reported residues in the 3D viewer).

Despite involving multiple external databases and full-text retrieval, the pipeline runs in real time. Discovery, download, and RAG construction typically complete within one to two minutes for a new protein, and subsequent queries on the same target are handled in seconds thanks to caching of the prebuilt index.

3.3 Agentic 3D Grounding and Interfaces with Scientists

Understanding proteins requires the navigation of multiple types of information, such as three-dimensional structures, experimental assay tables, and textual annotations. A central

goal of *Speak to a Protein* is to connect these diverse modalities through a single conversational interface, ensuring that system responses are not only generated in natural language but are also grounded in concrete evidence such as 3D visualizations, data tables, and executable code. This grounding operates bidirectionally: users can visually verify AI-generated hypotheses, while the AI agent itself can capture and interpret viewer states to inform its reasoning. To achieve this, we provide a set of interactive interfaces that extend dialogue into complementary domains: a structural viewer for molecular inspection, a tabular analysis environment for filtering and plotting assay data, and mechanisms for synchronizing actions across views. Together, these interfaces enable users to fluidly transition between asking questions, running analyses, and visually verifying hypotheses. Significantly, we are building these capabilities on top of a web application that has already attracted more than 18,000 registered users over the years, even in the absence of these AI features. These new capabilities enable medicinal chemists with no-code experience to use the tools like an expert computational chemist. We thus anticipate that it will be used by a considerable number of scientists, as demonstrated by the platform usage metrics in Appendix S1.4.

In *Speak to a Protein*, these functionalities are built using an entirely client-side sandbox for dynamic visualization and manipulation of molecular structures²¹. The sandbox builds on a browser-based molecular visualization toolkit that combines the high-performance `mol*` visualization engine²², capable of rendering large biomolecular structures and molecular dynamics trajectories directly in the browser, with a WebAssembly-enabled Python runtime (Pyodide). This allows the use of powerful Python libraries such as MoleculeKit¹⁹ in the client environment, enhancing the viewer’s capabilities to load and manipulate a wide range of common structural file formats, from PDB and CIF to molecular dynamics trajectory files such as XTC and TRR.

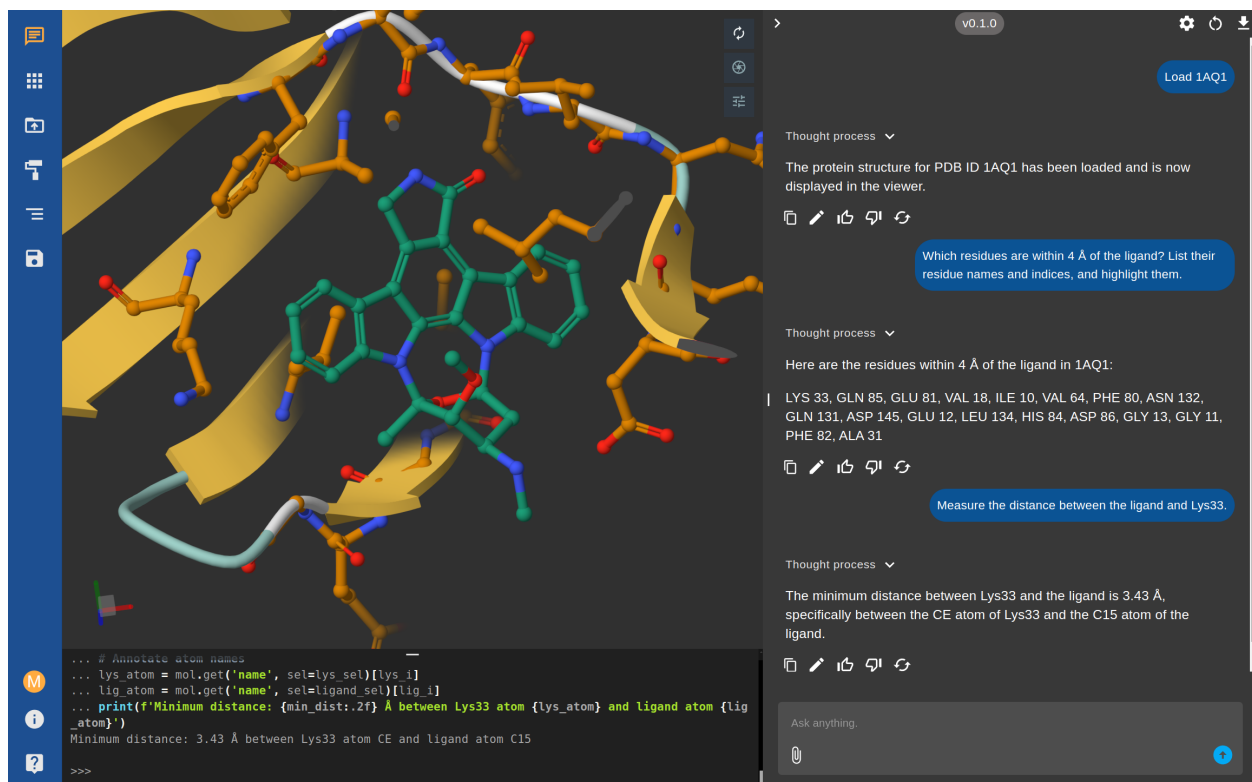


Figure 2: **Interactive analysis of the CDK2 structure (PDB: 1AQ1) using the *Speak to a Protein* multimodal assistant.** The user enters natural language queries in the chat panel (right), and the system responds with both textual answers and real-time updates to the 3D visualization (left). Python code executed in the integrated sandbox is shown in the console (bottom), providing full transparency and reproducibility of the underlying analyses.

A key feature is the ability to control this viewer through natural language commands. Requests such as "highlight the ATP-binding site" are translated into tool calls that execute Python code within the viewer, producing visual changes in real time (Figure 2). Beyond user-initiated actions, the agent can autonomously capture screenshots of the current viewer state and analyze them using vision capabilities. This agentic 3D vision enables the model to verify the results of its own manipulations, identify structural features that may not be explicitly encoded in the data, and iteratively refine visualizations based on what it observes.

4 Experiments

We present a set of illustrative case studies to show the capabilities and versatility of *Speak to a Protein*. Throughout all experiments reported in this section, we employ **GPT-5.2** as the foundational language model driving the *Speak to a Protein* agent.

Evaluating agentic systems for computational chemistry is challenging: existing benchmarks target either language understanding or isolated tasks, but few assess integrated reasoning over 3D structures, databases, and domain tools. We therefore evaluate on two complementary benchmarks: a *Simple Bio-Structural Benchmark* we developed to test basic computational chemistry capabilities; and *LiveProteinBench*²³, a contamination-free benchmark for protein property prediction. We also present two case studies using the dopamine D3 receptor (D3R)²⁴ and cyclin-dependent kinase 2 (CDK2)²⁵. These examples highlight how the system enables users to ask scientific questions, integrate data from multiple sources, and generate insights through interactive multi-modal analysis. We demonstrate the system’s ability to produce a structured summary report in LaTeX from all gathered information. By indexing this information, the system creates a queryable knowledge base that can be shared with other users.

4.1 Bio-Structural Benchmark

We established a benchmark of 20 tasks designed to test capabilities essential for computational chemistry. The dataset spans two categories: (i) **Structural Geometry** (e.g., calculating RMSD, detecting steric clashes, measuring residue distances); and (ii) **Sequence Retrieval** (e.g., extracting specific chain sequences). We compared *Speak to a Protein* against *GPT-5.2* with web search and Python, representative of the tools increasingly adopted by scientists for research tasks. While this baseline offers reasoning capabilities, web access, and code execution, it lacks specialized biochemical tools, a persistent file system for cross-referencing API outputs, and integrated 3D visualization. Tasks were scored as correct

or incorrect based on ground truth derived from programmatic calculations and database entries.

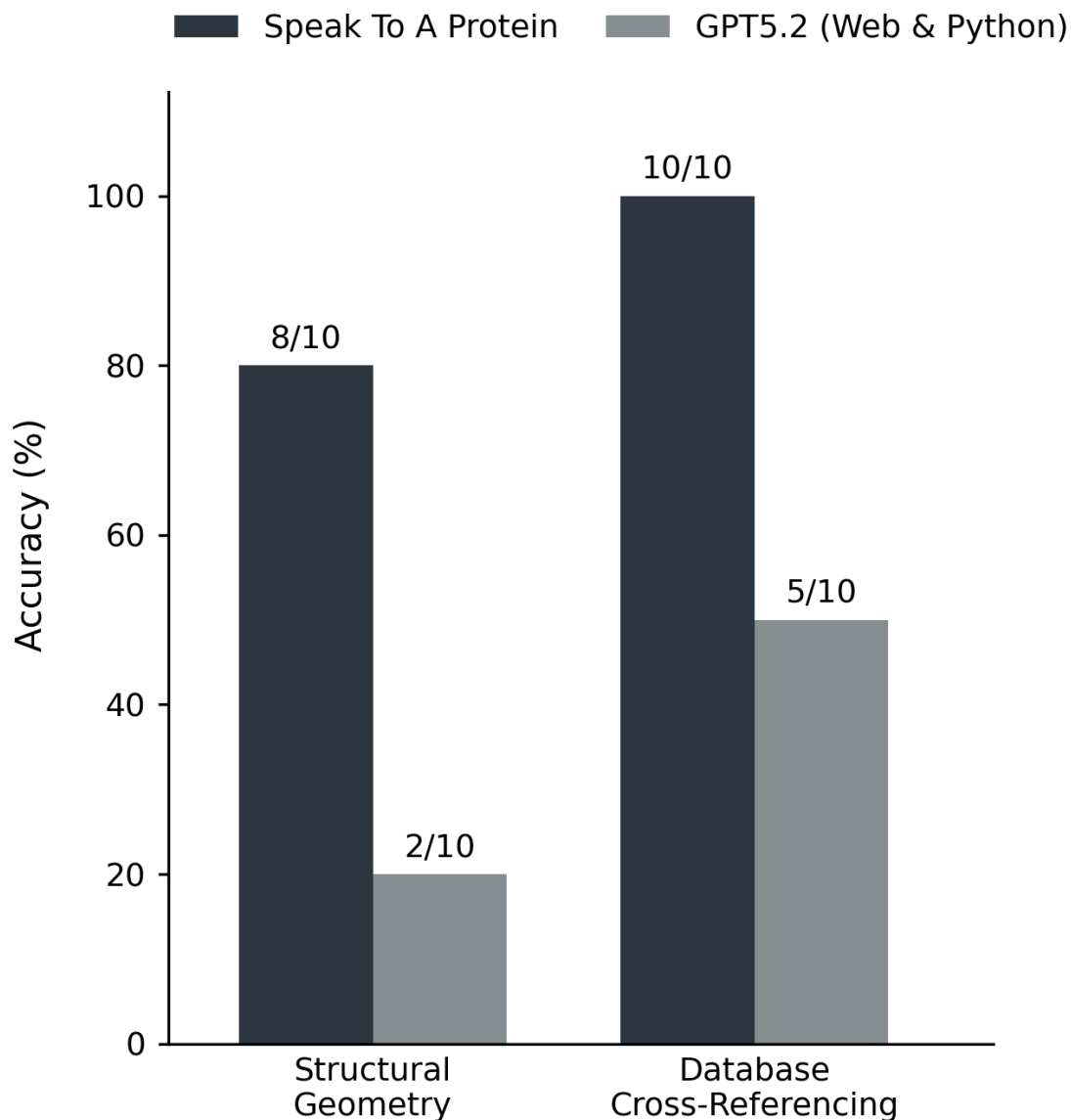


Figure 3: **Simple Bio-Structural Benchmark.** Accuracy by category for *Speak to a Protein* and *GPT-5.2 (Web & Python)* on 23 curated tasks.

As shown in Figure 3, *Speak to a Protein* achieved an accuracy of **90% (18/20)**, significantly outperforming the baseline at **35% (7/20)**. *GPT-5.2* frequently failed on tasks requiring precise structural geometry or specific bioactivity data. While it represents a

powerful general-purpose resource for scientists, it lacks the specialized infrastructure that computational chemists require. Its reliance on web search often resulted in retrieval errors, and without a specialized environment to manipulate and visualize molecular structures, it could not accurately derive geometric properties. In contrast, our system succeeded by executing analysis scripts and querying domain tools, ensuring that answers were derived directly from raw data and grounded in the interactive 3D viewer. The full list of benchmark questions is provided in Appendix S1.5.

4.2 Ablation study with LiveProteinBench

To assess the specific impact of our scientific data tools, we conducted an ablation study using LiveProteinBench²³, a contamination-free benchmark organized into three categories: physicochemical properties, structural localization, and functional annotation.

In this experiment, we disabled the scientific data retrieval tools of *Speak to a Protein*, restricting the system to its agentic 3D grounding and coding capabilities. Without access to specialized databases, *Speak to a Protein* achieved an overall accuracy of **53%**, performing similarly to GPT-5.2 at **51%** (Figure 4). This stands in stark contrast to the Bio-Structural Benchmark (Section 4.1), where our full system, equipped with all its tools, outperformed the baseline. This ablation study thus highlights that while 3D reasoning and code execution are vital for analysis, the scientific data tools are the primary drivers of domain-specific performance. Without them, the agent lacks the empirical evidence, such as the PDB structures themselves, necessary to ground its visual and computational capabilities.

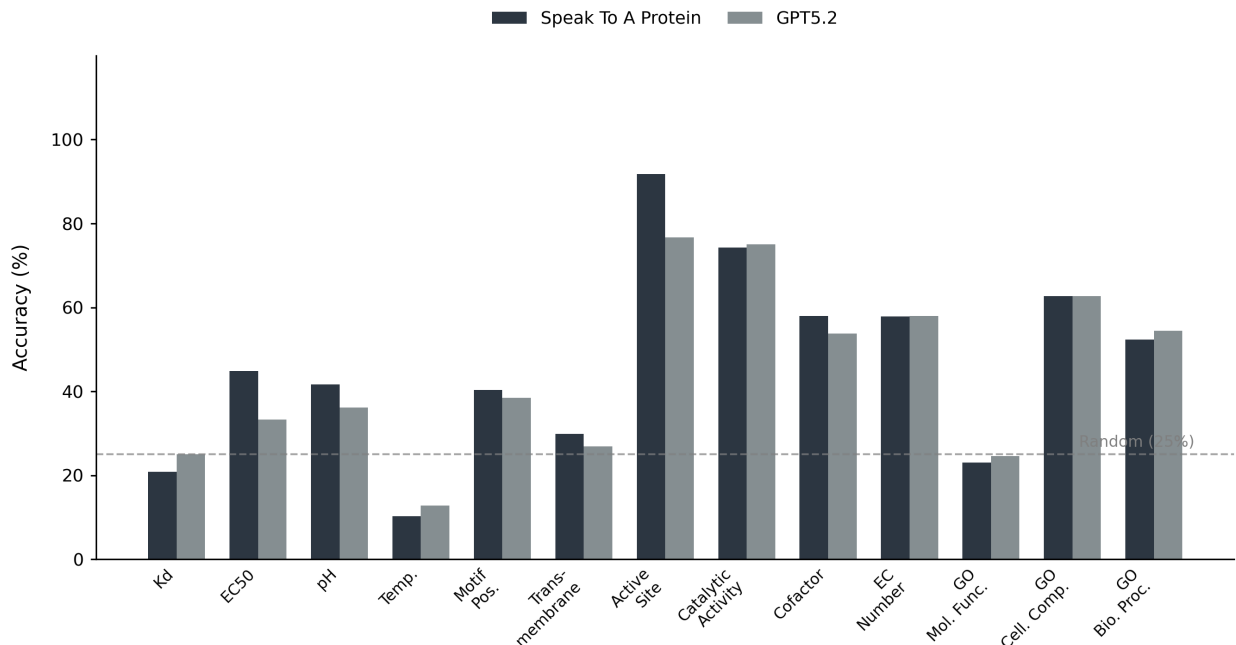


Figure 4: **LiveProteinBench Performance.** Accuracy by task for *Speak to a Protein* and GPT-5.2 across 12 protein property prediction tasks. Dashed line indicates random baseline (25%).

4.3 Speaking about the Dopamine D3 receptor (D3R)

We use *Speak to a Protein* to address a series of research questions related to D3R, a G protein-coupled receptor of significant pharmacological interest. In the video trace <https://youtu.be/H6ag4JJAM0w>, we show a possible user interaction with our system centered on D3R. The user begins by asking about the available structures for this receptor, loading the listed structure 3PBL. Next, the user instructs the system to filter for chain A, and changes the visual representation to focus on the binding pocket. Finally, it requests a list of known inhibitors associated with D3R. All this information is collected and made available to the AI so that knowledge is gathered and contextualized.

Focusing on the last query, we illustrate the workflow of the system (Figure 5). Upon the user’s request, the AI used several tools in sequence to produce the necessary data. Using these tools, it identified the correct UniProt entry for D3R and used it to query ChEMBL for all relevant bioactivity data. The retrieved assay results were compiled and automatically

stored in a CSV file, which was then loaded directly into the viewer for exploration and analysis. The resulting table provides an overview of all known D3R inhibitors, along with chemical structures, assay details, potency metrics (such as EC_{50} , IC_{50} , K_i), and references to the corresponding literature. Notable examples include inhibitors with subnanomolar to low nanomolar potencies such as CHEMBL5841759 (K_i : 0.012 nM) and CHEMBL5802711 (K_i : 0.014 nM). The results also listed several potent reference agonists for context.

The figure displays a software interface for exploring D3R inhibitors. On the left, a table lists inhibitors with their chemical structures, ChEMBL IDs, standard units, and standard values. On the right, a chat panel shows a user query and the system's response, including a high-level plan and the tools used.

canonical smiles	molecule chembl id	standard units	standard value
	CHEMBL5841759	nM	0.012
	CHEMBL5841759	nM	0.012
	CHEMBL5802711	nM	0.014
	CHEMBL5918571	nM	0.014

Chat Panel:

User query: Give me all the known inhibitors for D3R and their affinities. Load the resulting table.

Thought process:

High-Level Plan:

1. Identify the correct target entry for D3R in major bioactivity databases (ChEMBL).
2. Retrieve all assay data (focusing on inhibitors and affinities) for D3R.
3. Store the result in a CSV file for easy inspection.
4. Load the CSV into the Viewer for direct analysis.

Used tool: uniprot_text_search
 Used tool: uniprot_data_lookup
 Used tool: chembl_assays
 Used tool: pmview_load_file

All known inhibitors (and agonists/antagonists) for the human D3 dopamine receptor (D3R) and their affinities from functional assays have been retrieved from ChEMBL and loaded as a table in your Viewer. This includes the compound structures, assay details, potency metrics (e.g., EC_{50} , IC_{50} , K_i), and literature references.

Figure 5: **Retrieval and exploration of potent D3R inhibitors.** Example user query and system workflow for fetching all known D3R inhibitors and their affinities using *Speak to a Protein*. The chat panel shows the AI’s reasoning, including the tools used and the final response. On the left, the interface displays an interactive table of D3R inhibitors, including chemical structures, ChEMBL IDs, assay details, and more, enabling direct exploration and further analysis.

In the video trace <https://youtu.be/nER3vC90y1Q>, we show how to investigate the differences between the D3 and D2 receptors. The literature search capability can help rapidly surface and synthesize expert knowledge from primary sources to address detailed structural questions. Upon receiving the prompt “Based on the literature, compare the binding pockets

of D3R and D2R and summarize the main structural features that could be exploited for ligand selectivity.”, the system first uses *UniProt Search* to identify the canonical UniProt IDs for the specified proteins, ensuring precise target selection. Then, using *Literature Search*, an embedding-based literature search is performed, retrieving relevant articles and review information from PubMed Central that address the requested topic.

In this example, the system processed a set of 12 literature passages for D2R and 10 for D3R, selecting at least four distinct, peer-reviewed structural biology studies and related supporting works^{24,26–28}. Based on this, the generated answer highlighted that while the orthosteric binding pocket is highly conserved, selectivity can be achieved by targeting differences in the architecture and flexibility of the ‘extended binding pocket’, as well as in the conformation of extracellular loops. It also described the significance of distinct residues (such as Trp100 and residues in EL1/EL2, positions 1.39 and 7.35) that shape ligand binding modes and selectivity opportunities. The resulting analysis revealed that while D3R and D2R share a conserved binding pocket core, D2R possesses additional flexible and hydrophobic residues that create a deeper, more accommodating pocket. These structural differences, clearly visualized and annotated in the viewer, help explain how each receptor achieves ligand selectivity, providing actionable insights for targeted drug design.

4.4 Speaking about the Cyclin-dependent kinase 2 (CDK2)

We present a complete drug discovery workflow using cyclin-dependent kinase 2 (CDK2), a validated cancer target with extensive structural and bioactivity data²⁵. This example showcases how *Speak to a Protein* can streamline the entire process from initial target evaluation to actionable insights. Our analysis begins with a systematic exploration of available structural data, progresses through mining and filtering of bioactivity data, and culminates in an integrated structural-activity analysis. The workflow highlights the system’s ability to seamlessly transition between different data modalities and analysis types, all through natural language interactions.

Structure-activity. The complete conversational trace for this analysis is provided in Appendix S1.3. First, we ask the AI system to retrieve all available CDK2 structures from the Protein Data Bank. The system first used the UniProt tool to identify the canonical human CDK2 entry (P24941) and retrieve all 462 associated PDB structures. It then systematically queried each structure using the PDB information tool, which parsed each entry to extract co-crystallized ligands. After filtering out common solvents and ions, this process identified 479 unique ligand-structure pairs containing small molecules relevant for drug discovery. The system then automatically determined bioactivity coverage, revealing that 132 out of 258 ChEMBL-annotated ligands had experimental activity measurements available. Through systematic data cleaning and deduplication focused on IC₅₀ values, we generated a refined dataset of approximately 100 unique CDK2-ligand complexes, ranked by potency. The top 20 most potent complexes (IC₅₀ values ranging from sub-nanomolar to 15 nM) were loaded into the 3D viewer and structurally aligned. For detailed binding site analysis, the system identified and visualized only the ATP-binding pocket residues (within 6 Å of the co-crystallized ligands), excluding solvents and common crystallization agents (see Figure S1 in Appendix S1.2).

Automated Report Generation. Furthermore, the AI extracted the binding pocket sequences and stored them in FASTA format. Pairwise sequence alignment revealed high conservation across the ATP-binding site, with only minor variations at peripheral positions. The AI system then conducted a comprehensive literature search to contextualize these findings with existing CDK2 research, automatically generating a markdown summary of relevant studies discussing binding site features and structure-activity relationships. The complete report, automatically generated by the AI system synthesizing all gathered data into actionable insights suitable for distribution to medicinal chemistry teams, is provided in Appendix S1.2. The system’s ability to generate publication-ready reports and comprehensive literature summaries further enhances its utility in collaborative drug discovery

environments, where rapid communication of complex structural and biochemical insights is essential for decision-making.

5 Conclusion and Limitations

This study demonstrates how *Speak to a Protein* compresses traditional workflows involving several hours of manual data gathering, analysis, and synthesis into an interactive session taking less than an hour. The system’s ability to generate publication-ready reports further enhances its utility in collaborative drug discovery environments where rapid understanding and communication of complex structural and biochemical insights are essential for decision-making.

We found the following limitations in the current version of *Speak to a Protein*. The open web application accesses only public information, such as literature, structure-activity relationships, and structural data. This limits the data access and the understanding and downstream calculations. We plan to easily extend it with additional tools to access internal or proprietary datasets, further broadening the scope of information that can be queried and integrated. The 3D viewer can experience performance issues when rendering a large number of complex structures simultaneously. We also observe occasional difficulties in seamlessly connecting outputs between tools, stemming from different data representations across the system’s components, e.g., residue indices across literature and structural databases. Furthermore, long tool outputs can strain the model’s context window. Future work will focus on offloading large data payloads to files and equipping the model with a broader set of tools for file management, allowing for more robust and complex analytical workflows.

Impact Statement

This paper presents work whose goal is to advance the field of AI-assisted scientific discovery in structural biology and drug discovery. The system is designed to democratize access to

complex protein analysis by enabling researchers without specialized computational skills to perform advanced structural and biochemical analyses. Potential positive impacts include accelerating drug discovery timelines, reducing barriers to entry for bench scientists, and enabling more rapid hypothesis generation in biomedical research. We do not foresee specific negative societal consequences that must be highlighted, as the system operates on publicly available scientific data and is intended to augment, not replace, human scientific judgment.

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